MARITIME

IMO Polar Code
ARCTIC SHIPPING BEST PRACTICE FORUM

Experience so far…

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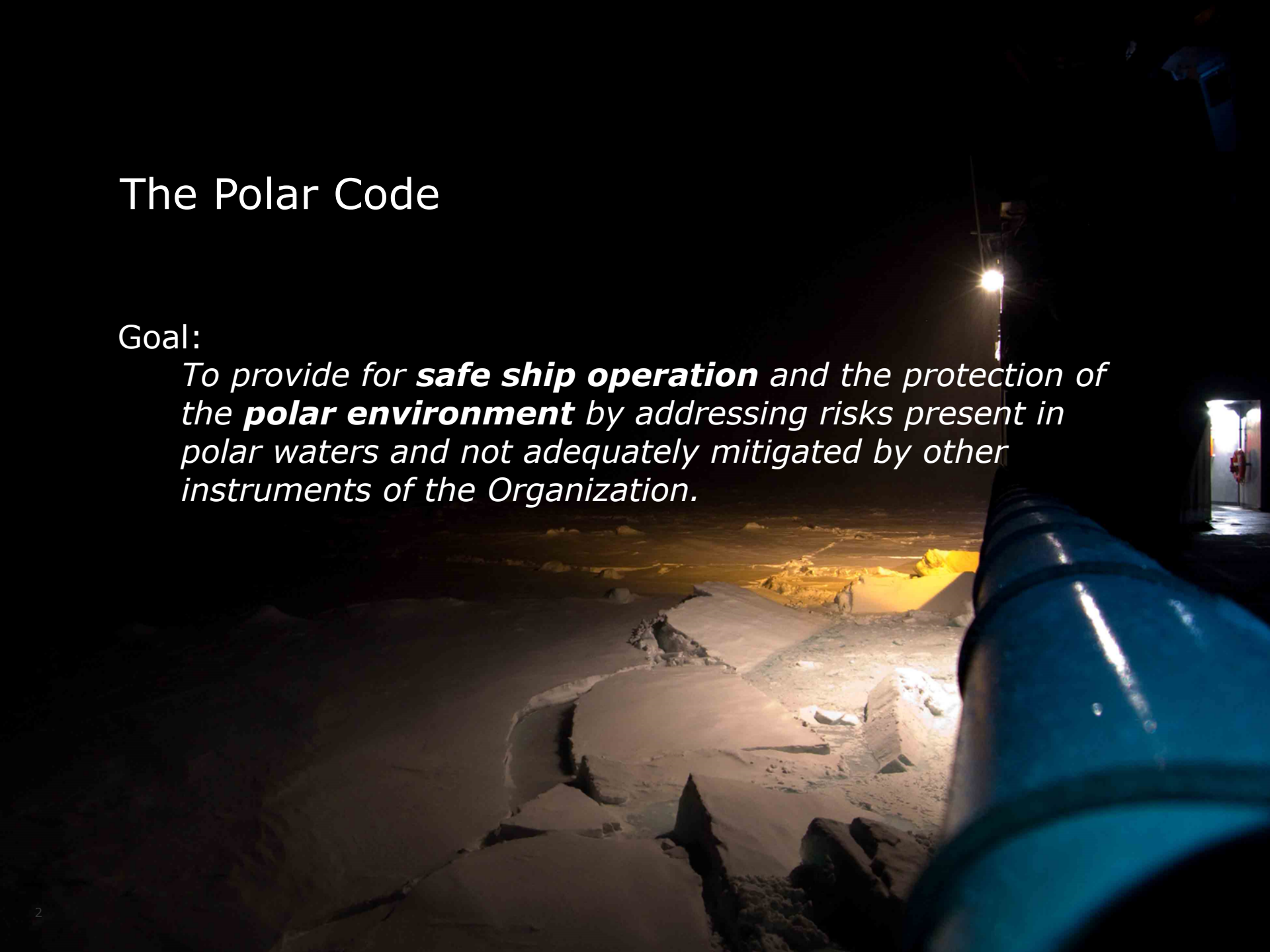
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https://www.dnvgl.com/maritime/polar/services.html

The Polar Code

Goal:

*To provide for **safe ship operation** and the protection of the **polar environment** by addressing risks present in polar waters and not adequately mitigated by other instruments of the Organization.*

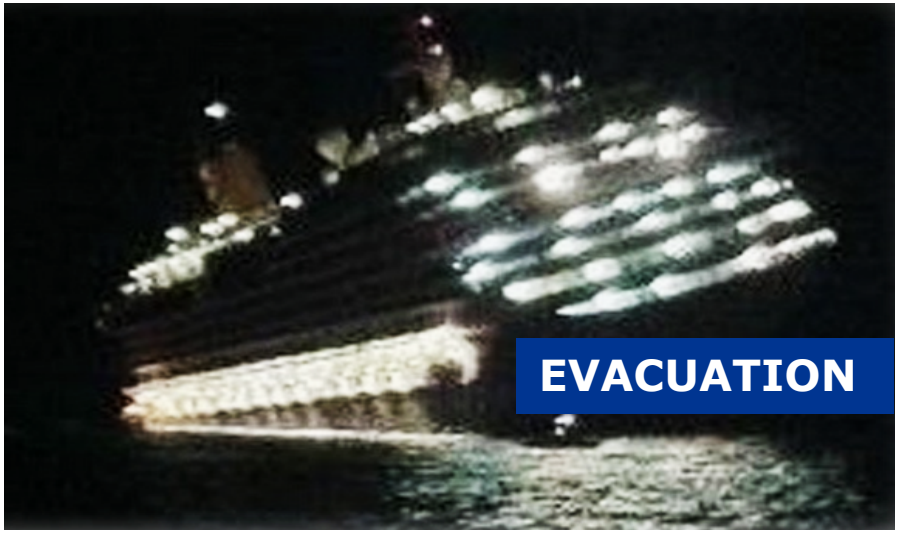


Challenges related to ship operations in Arctic waters









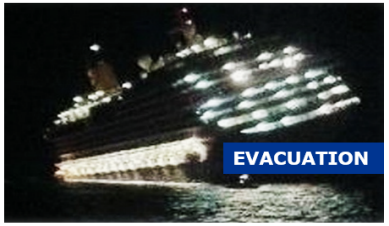
Risk = Probability x Consequence

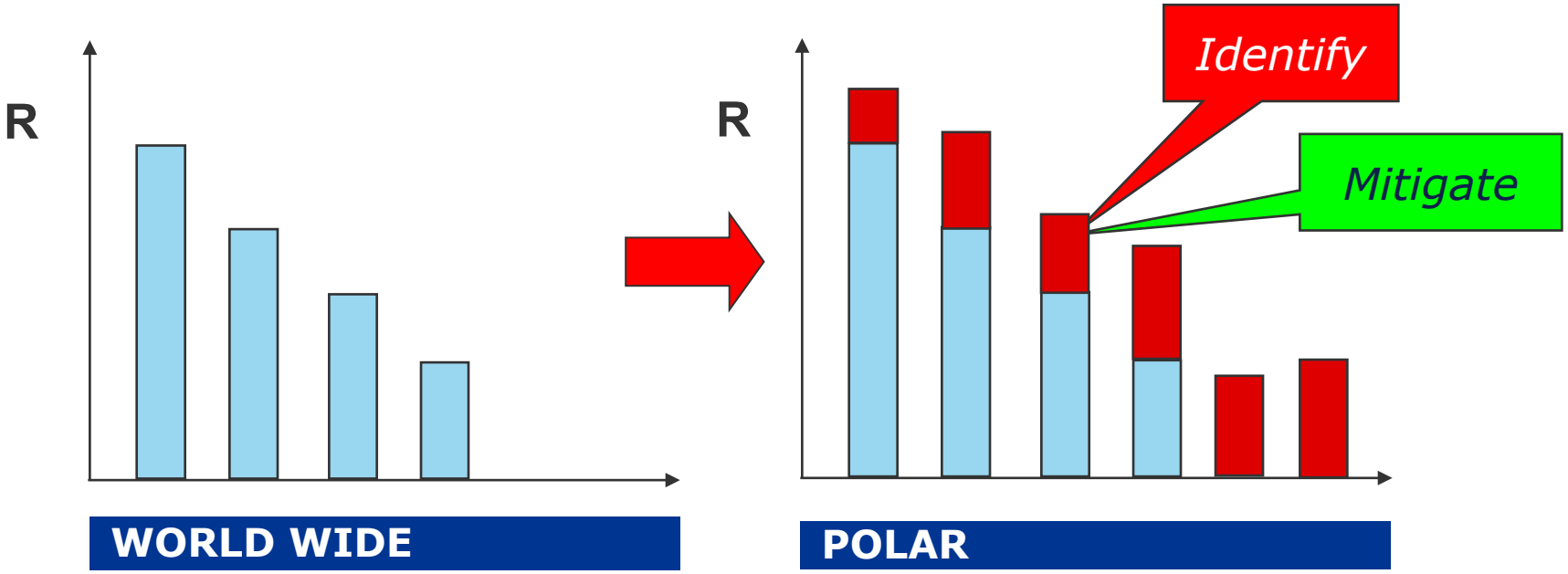












The main impacts of the Code; - additional requirements



- Operational (Risk) Assessment*
- PWOM, Polar Water Operational Manual*
- Life Saving Equipment
- Route planning
- Environment; no discharge = stricter
  - Requirements to holding tanks (volume), garbage storage etc.

- Finally, a Polar Ship Certificate is required and will be issued by DNV GL on behalf of flag

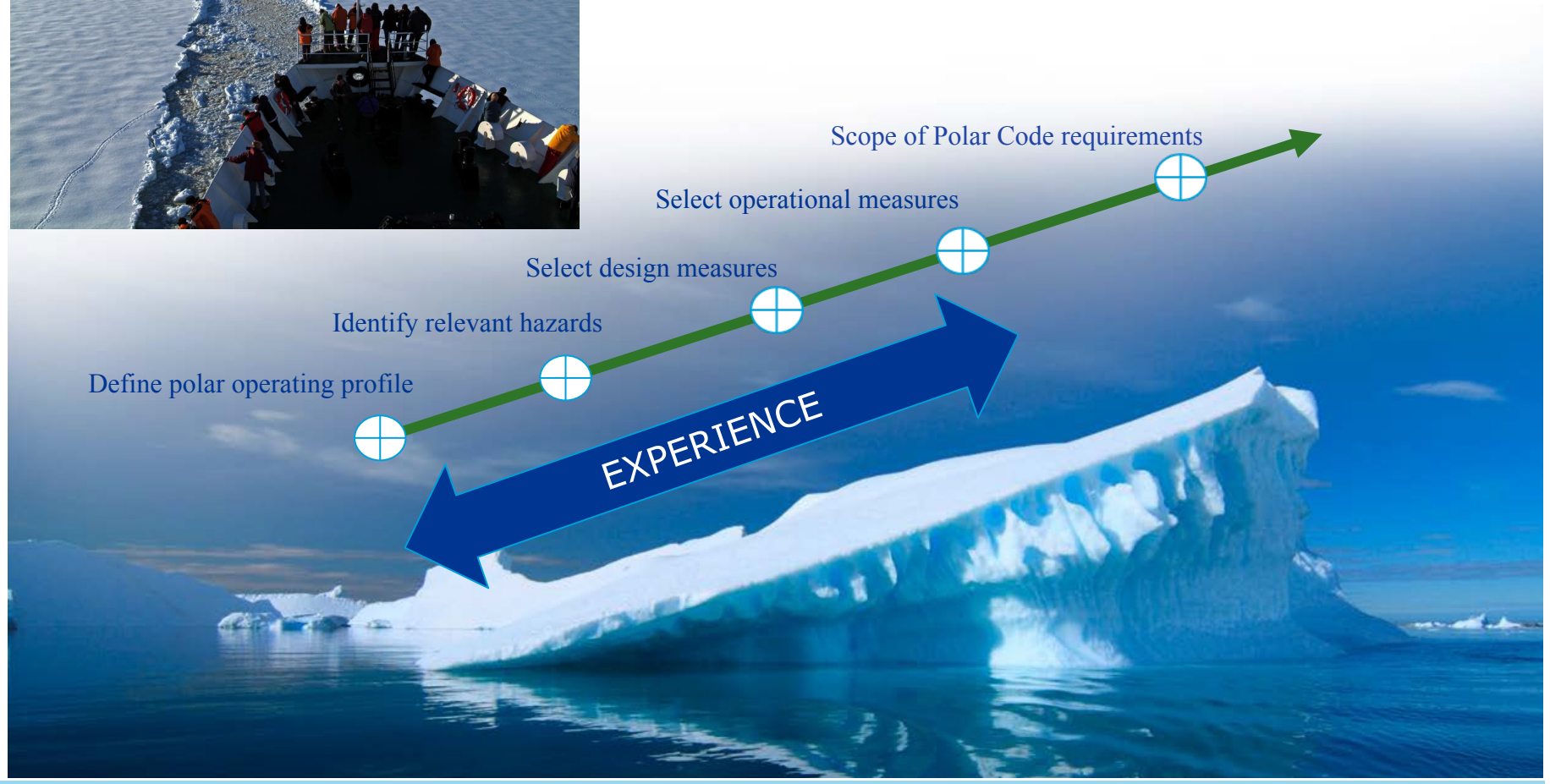
**Documents to be produced by owner and approved by DNV GL*

Table of content; Part IA SAFETY MEASURES

- Chapter 1: General
- Chapter 2: Polar Water Operation Manual
- Chapter 3: Ship Structure
- Chapter 4: Stability and Subdivision
- Chapter 5: Watertight and Weathertight Integrity
- Chapter 6: Machinery Installations
- Chapter 7: Fire Safety/Protection
- Chapter 8: Life Saving Appliances and Arrangement
- Chapter 9: Safety Navigation
- Chapter 10: Communication
- Chapter 11: Voyage Planning
- Chapter 12: Manning and Training Familiarity
- Part IB: Additional guidance regarding the provisions of the introduction and Part IA

Operational Assessment, the first step





Experience from:

- **Chapter 1 General**
  - **Operational Assessment**
  - Include all additional HAZARDS
  - Example on table to be used in OA:



ITEM	EQUIPMENT	HAZARD	R = P x C	RISK CONTROL ACTIONS	STATUS	COMMENT
					OK	
					More info	
					No Go	
				Design, Equipment or Operational procedure (->PWOM)		

Experience from:

- **Chapter 2: Polar Water Operation Manual**
  - Different interpretations, ref. chapt. 2 and Appendix 2.
  - Limited understanding of the main goal of the manual
  - Need for a **template** for table of content
  - The connection between the Operational Assessment and PWOM not understood
- **Chapter 3: Ship Structure**
  - Relation between Categories and Ice Class.
  - Actual ice limit for Cat. C is maximum thin firs year ice (0.3m)
  - Category refers to ice class but not same definition of ice conditions (WMO)
- **Chapter 4: Stability and Subdivision**
  - Confusion if damaged stability (Cat. A and B) requirements to be fulfilled with ice accretion

Experience from:

- **Chapter 5: Watertight and Weathertight Integrity**
  - ok
- **Chapter 6: Machinery Installations**
  - ok
- **Chapter 7: Fire Safety/Protection**
  - ok
- **Chapter 8: Life Saving Appliances and Arrangement**
  - Maybe the 5-days requirement to equipment should be possible to reduce based on actual operation. Example, OSV during field operation and cruise vessels with limited operations
  - Gap between actual requirements to equipment to ensure survival and what will be required by flags. (LB capacity, use of rafts, possible de-rating etc.)
  - Content and quality of PSK and GSK
  - Thermal protection
  - TPA and survival suites

Experience from:

- **Chapter 9: Safety Navigation**
  - Definition of up-to-date information including ice information (9.3.1)
- **Chapter 10: Communication**
  - Battery capacity/operational procedure
  - TMAS (need satellite communication, often not approved solutions) ref. (10.3.1.4)
- **Chapter 11: Voyage Planning**
  - ok
- **Chapter 12: Manning and Training Familiarity**
  - Actual training requirements, - how to fulfil?
  - With regard to training and STCW, the following Implementation timeline has to be followed:
    - The amendments to the STCW enter into force on 1 July 2018. Implementation of them includes a two-year transition period (from 2018 to 2020). Ship owners and seafarers should contact their national maritime administration for information on how it is implementing the Polar Code training requirements and how to obtain the necessary certificates.
  -

The process towards receiving the certificate

Owners Polar Operating Profile

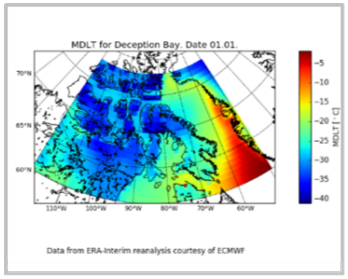
Operational Assessment

For Information

Polar Water Operational Manual, PWOM

Approved

Survey -> Certificate



Item No.	Equipment	Equipment details	Hazard/Probability	Consequence	Risk control actions	Status	Comments
						OK	
						OK	
						Additional Measures	
						NO GO	





Some general challenges

- Testing procedures for equipment, for PST lower than -25° C
- Data/picture communications required for TMAS and Ice information
  - Equipment to satisfy same Performance Standard referred to in SOLAS
- Actual need for GSK, - if evacuation to land/ice
- If OA concludes with dry evacuation, TPA ok and no need for survival suites
  - Definition of Thermal Protective Aid? =TPA?
- PSC: Definition of Operational Limitations; Ice Conditions
  - Refer to Polaris?
  - What about vessels with documented experience in more severe ice than the Category definition?
  - Info about possible operation in darkness and exposure to icing may be included.



DNV GL have so far more than 60 vessels approved or in the loop



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